Functions, Rational Expressions & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Simplifying Rational Expressions & Domain Restrictions

Factor the numerator and denominator completely, state all domain restrictions, and simplify $\frac{2x^2 - 5x - 12}{x^2 - 16}$.

2 [Foundational] • [FRQ] • Target: Adding & Subtracting Rational Expressions

Combine over a common denominator and simplify into a single reduced fraction: $\frac{3}{x-2} - \frac{2x-1}{x^2-4}$.

3 [Foundational] • [FRQ] • Target: Solving Rational Equations

Solve the rational equation $\frac{4}{x-1} + \frac{2}{x+3} = 1$ for all valid real solutions.

4 [Foundational] • [FRQ] • Target: Complex Number Arithmetic

Perform each complex number operation and write the result in standard form $a + bi$:

a $(5 - 3i)(2 + 4i)$

b $(3 - 2i)^2 - (4 + 5i)$

5 [Foundational] • [FRQ] • Target: Complex Conjugates & Division

Divide the complex numbers $\frac{14 - 5i}{3 + 2i}$ by multiplying numerator and denominator by the complex conjugate, and express the result in the form $a + bi$.

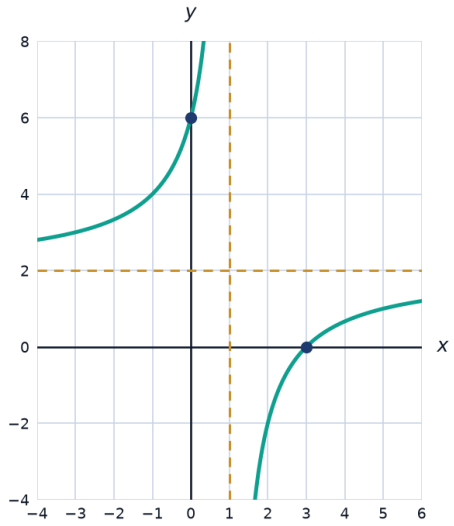
6 [Foundational] • [FRQ] • Target: Powers of the Imaginary Unit i

Simplify $i^{45} + i^{102} - i^{79}$ using the periodicity of powers of i ($i^4 = 1$).

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7 [Foundational] • [FRQ] • Target: Asymptotes & Intercepts of Rational Functions

State the vertical asymptote, horizontal asymptote, x-intercept, and y-intercept of the rational function $f(x) = \frac{2x-6}{x-1}$ graphed below.



Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Extraneous Solutions in Rational Equations

Solve the rational equation $\frac{x}{x-3} - \frac{3}{x-3} = 2$ and explain why it has no solution.

9 [Intermediate] • [FRQ] • Target: Partial Fraction Decomposition

Find the constants A and B in the partial-fraction identity $\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.

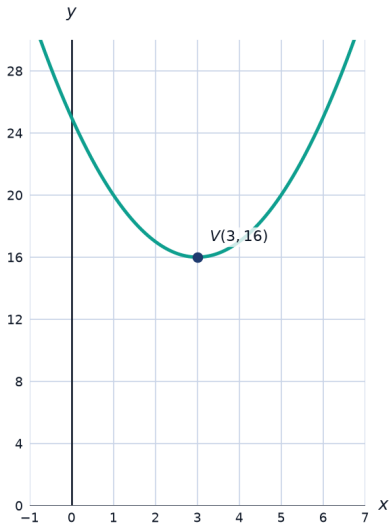
10 [Intermediate] • [FRQ] • Target: Simplifying Complex Fractions

Simplify the complex fraction $\frac{\frac{1}{x}-\frac{1}{3}}{x-3}$ for $x \neq 0, 3$, and evaluate the simplified expression at $x = 3$.

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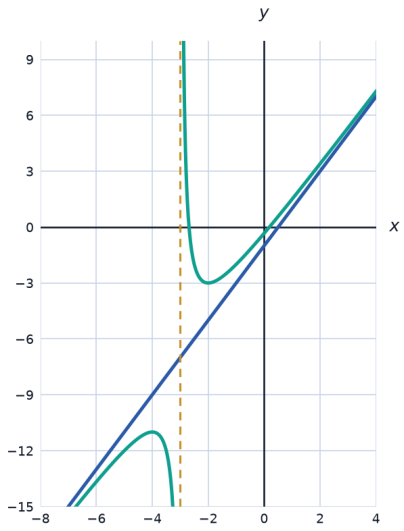
11 [Intermediate] • [FRQ] • Target: Complex Roots of Quadratic Equations

Find the complex conjugate solutions to $x^2 - 6x + 25 = 0$ in the form $a \pm bi$, and verify that their sum is 6 and their product is 25.



12 [Intermediate] • [FRQ] • Target: Quotient-Remainder Form of Rational Functions

Use polynomial long division to rewrite $\frac{2x^2 + 5x - 1}{x + 3}$ in the form $ax + b + \frac{r}{x + 3}$.



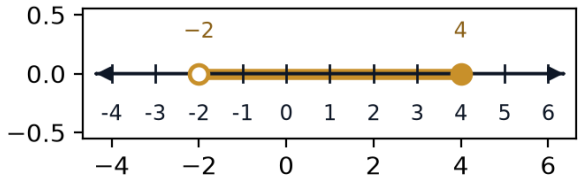
13 [Intermediate] • [FRQ] • Target: Harmonic / Parallel Resistance Formulas

Two resistors R_1 and R_2 connected in parallel have equivalent resistance R_{eq} given by $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$. Solve for R_2 in terms of R_{eq} and R_1, and compute R_2 when $R_{\text{eq}} = 12\,\Omega$ and $R_1 = 20\,\Omega$.

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14 [Intermediate] • [FRQ] • Target: Solving Rational Inequalities

Solve the rational inequality $\frac{x-4}{x+2} \leq 0$ and express the solution set in interval notation.



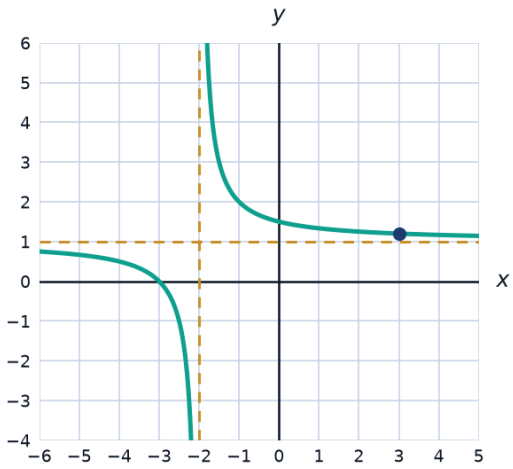
Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Complex Conjugate Root Theorem • Cross-Lesson: Topic 4.01 (Cubic Polynomials)

A polynomial $P(x)$ of degree 3 with real coefficients has zeros $x = 2$ and $x = 1 + 3i$, and satisfies $P(0) = -40$. Find the factored and expanded forms of $P(x)$.

16 [Challenging] • [FRQ] • Target: Removable Discontinuities vs. Vertical Asymptotes • Cross-Lesson: Topic 3.01 (Quadratic Factoring)

Find the domain, removable discontinuity (hole) coordinates, and vertical asymptote of $g(x) = \frac{x^2 - 9}{x^2 - x - 6}$.



17 [Challenging] • [FRQ] • Target: Upstream-Downstream Rational Modeling • Cross-Lesson: Topic 5.01 (Rates) & Topic 3.01 (Quadratics)

A kayaker paddles 12 km upstream and 12 km back downstream in a river whose current flows at 3 km/h. If the total round-trip time is 3 hours, set up a rational equation to find the kayaker's speed v in still water.

18 [Challenging] • [FRQ] • Target: Inverting Linear Fractional Transformations • Cross-Lesson: Topic 1.01 (Isolating Variables)

For a one-to-one rational function $f(x) = \frac{3x + 2}{x - 4}$ $(x \neq 4)$, find the formula for its inverse function $f^{-1}(x)$ and state the domain and range of f.

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19 [Challenging] • [FRQ] • Target: Modulus Algebra in the Complex Plane • Cross-Lesson: Topic 7.01 (Pythagorean Theorem)

In the complex plane, find the modulus $|z|$ of $z = \frac{(3+4i)(1-i)}{2-2i}$ without expanding the numerator.

20 [Challenging] • [FRQ] • Target: Rational-Reciprocal Identities & Optimization • Cross-Lesson: Topic 3.01 (Completing the Square)

If $x > 0$ and $x + \frac{4}{x} = 6$, find the exact value of $x^2 + \frac{16}{x^2}$ and the minimum possible value of $t + \frac{4}{t}$ for all $t > 0$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Simplifying Rational Expressions

Given the algebraic relationship in which for $x \neq 3$, which expression is equivalent to $\frac{x^2-9}{x-3}$?

a. $x - 3$

b. $x + 3$

c. $x^2 + 3$

d. $x - 9$

22 [Foundational] • [MCQ] • Target: Adding Complex Numbers

If $i = \sqrt{-1}$, what is the sum $(7 - 4i) + (-2 + 9i)$?

a. $5 + 5i$

b. $5 - 13i$

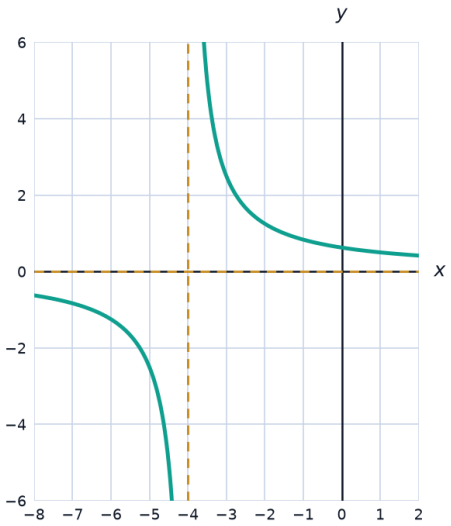
c. $9 + 5i$

d. $9 - 5i$

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23 [Foundational] • [MCQ] • Target: Vertical Asymptotes

What is the vertical asymptote of the graph of $f(x) = \frac{5}{2x+8}$?



a. $x = -4$

b. $x = 0$

c. $x = 4$

d. $x = 5$

24 [Foundational] • [MCQ] • Target: Proportions & Rational Equations

Within the real coordinate domain, to satisfy the synthesis condition, if $\frac{6}{x-2} = \frac{3}{4}$, what is the value of x?

a. 6

b. 8

c. 10

d. 12

25 [Foundational] • [MCQ] • Target: Product of Complex Conjugates

What is the product $(4 + 3i)(4 - 3i)$?

a. 7

b. $16 - 9i$

c. 25

d. $7 + 24i$

26 [Foundational] • [MCQ] • Target: Adding Fractions with Variable Denominators

Combine the rational terms $\frac{2}{x} + \frac{5}{3x}$ $(x \neq 0)$ into a single simplified fraction.

a. $\frac{7}{4x}$

b. $\frac{7}{3x}$

c. $\frac{11}{3x}$

d. $\frac{10}{3x^2}$

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27 [Foundational] • [MCQ] • Target: Powers of i

What is the value of i^{26}?

a. -1

b. 1

c. $-i$

d. i

28 [Foundational] • [MCQ] • Target: Domain of Rational Functions

For what value of x is the function $h(x) = \frac{x+5}{x^2-6x+9}$ undefined?

a. -5

b. -3

c. 0

d. 3

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Rewriting Improper Rational Expressions

Rewrite the improper rational function $\frac{3x+7}{x+2}$ $(x \neq -2)$ in quotient-plus-remainder form.

a. $3 + \frac{1}{x+2}$

b. $3 + \frac{7}{x+2}$

c. $3 + \frac{5}{x+2}$

d. $3 - \frac{1}{x+2}$

30 [Intermediate] • [MCQ] • Target: Complex Division & Real/Imaginary Parts

Analyze the configuration in which when $\frac{10}{1-2i}$ is written in the form $a + bi$, what is the value of $a + b$?

a. 2

b. 4

c. 6

d. 8

31 [Intermediate] • [MCQ] • Target: Filtering Extraneous Roots

What are all real solutions to $\frac{x^2}{x-4} = \frac{16}{x-4}$?

a. $x = 4$ only

b. $x = -4$ only

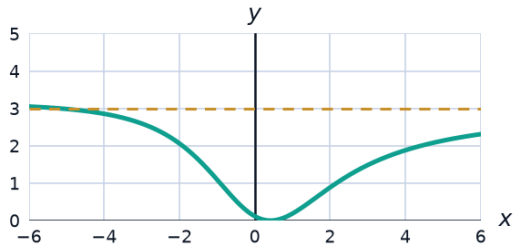
c. $x = 4$ and $x = -4$

d. No real solution

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32 [Intermediate] • [MCQ] • Target: Horizontal Asymptotes of Equal-Degree Rationals

What is the horizontal asymptote of $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 9}$?



a. $y = 0$

b. $y = \frac{1}{3}$

c. $y = 3$

d. $y = 6$

33 [Intermediate] • [MCQ] • Target: Partial Fractions Cover-Up Method

If $\frac{4x - 2}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}$ for all $x \neq \pm 1$, what is the value of A?

a. 1

b. 2

c. 3

d. 4

34 [Intermediate] • [MCQ] • Target: Simplifying Compound Fractions

Simplify the compound reciprocal fraction $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{ab}}$ for positive real numbers $a, b > 0$.

a. $\frac{1}{a + b}$

b. $a + b$

c. ab

d. $\frac{a + b}{a^2 b^2}$

35 [Intermediate] • [MCQ] • Target: Equating Real & Imaginary Parts

If $(x + yi)(2 - i) = 8 + i$, where x and y are real numbers, what is the value of $x + y$?

a. 3

b. 4

c. 5

d. 6

36 [Intermediate] • [MCQ] • Target: Rational Expressions for Combined Work Rates

Printer A can print a batch of reports in x hours, and Printer B can print the same batch in $x + 3$ hours.

Which expression represents the fraction of the batch they can print together in 1 hour?

a. $\frac{2}{2x + 3}$

b. $\frac{2x + 3}{x(x + 3)}$

c. $\frac{x(x + 3)}{2x + 3}$

d. $\frac{1}{x(x + 3)}$

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37 [Intermediate] • [MCQ] • Target: Reconstructing Quadratics from Complex Conjugate Roots

Which of the following quadratic equations has roots $2 + i$ and $2 - i$?

a. $x^2 - 4x + 3 = 0$

b. $x^2 - 4x + 5 = 0$

c. $x^2 + 4x + 5 = 0$

d. $x^2 - 2x + 5 = 0$

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Periodic Series of Powers of i • Cross-Lesson: Topic 4.01 (Exponents & Periodicity)

What is the value of the sum $i^1 + i^2 + i^3 + \ldots + i^{99}$?

a. -1

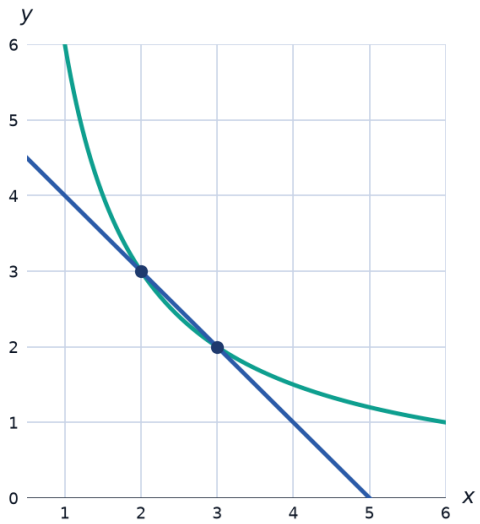
b. 0

c. i

d. $-i$

39 [Challenging] • [MCQ] • Target: Hyperbola-Line Intersection System • Cross-Lesson: Topic 2.01 (Systems) & Topic 3.01 (Quadratics)

In the xy-plane, at how many points does the rational curve $y = \frac{6}{x}$ intersect the line $x + y = 5$?



a. Zero

b. One

c. Two

d. Three

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40 [Challenging] • [MCQ] • Target: Rational Dilution Modeling & Asymptotic Behavior • Cross-Lesson: Topic 5.01 (Mixture & Concentration)

A chemistry student adds x liters of pure water to 20 liters of a 40% acid solution. Which rational function gives the new acid concentration $C(x)$ as a decimal, and what is $\lim_{x\to\infty} C(x)$?

a. $C(x) = \frac{8}{20+x}$, horizontal asymptote $C = 0$

b. $C(x) = \frac{8+x}{20+x}$, horizontal asymptote $C = 1$

c. $C(x) = \frac{0.40}{20+x}$, horizontal asymptote $C = 0$

d. $C(x) = \frac{20}{8+x}$, horizontal asymptote $C = 0$

41 [Challenging] • [MCQ] • Target: Integer Divisibility via Rational Division • Cross-Lesson: Topic 4.01 (Polynomial Division)

For how many integer values of n is $\frac{2n+15}{n+2}$ also an integer?

a. 2

b. 4

c. 6

d. 8

42 [Challenging] • [MCQ] • Target: 90-Degree Rotation by i in the Complex Plane • Cross-Lesson: Topic 7.01 (45-45-90 Right Triangles)

If $z = 3 + 4i$, what is the distance in the complex plane between z and iz?

a. 5

b. $5\sqrt{2}$

c. 7

d. 10

43 [Challenging] • [MCQ] • Target: Non-Zero Right-Hand-Side Rational Inequalities • Cross-Lesson: Topic 2.01 (Inequalities)

What is the solution set to the rational inequality $\frac{x+5}{x-2} \geq 2$?

a. $(2, 9]$

b. $[9, \infty)$

c. $(-\infty, 2) \cup [9, \infty)$

d. $[-5, 2)$

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44 [Challenging] • [MCQ] • Target: Inverse Pythagorean Theorem for Right-Triangle Altitude • Cross-Lesson: Topic 7.01 (Right-Triangle Altitude)

A right triangle has legs of lengths $\frac{1}{a}$ and $\frac{1}{b}$, where $a, b > 0$. Which rational expression gives the length of the altitude drawn to the hypotenuse?

a. $\frac{1}{a+b}$

b. $\frac{1}{\sqrt{a^2+b^2}}$

c. $\frac{ab}{\sqrt{a^2+b^2}}$

d. $\frac{\sqrt{a^2+b^2}}{ab}$

45 [Challenging] • [MCQ] • Target: Iterated Composition of Rational Functions • Cross-Lesson: Topic 4.01 (Function Composition)

If $f(x) = \frac{x-1}{x+1}$, what is $f(f(x))$ for all $x \neq -1, 0$?

a. x

b. $-\frac{1}{x}$

c. $\frac{1}{x}$

d. $-x$

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